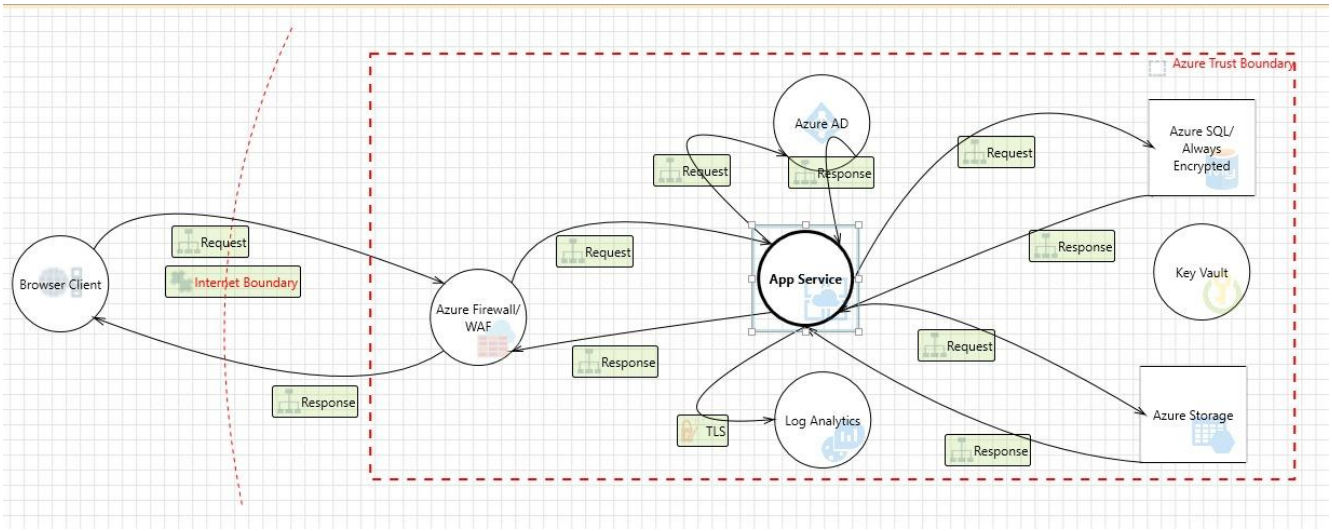
LABORATORIO AZURE PC4 — WAF, ENTRA ID, LOG ANALYTICS Y ALWAYS ENCRYPTED

#Curso: CIB59-M | Seguridad en la Nube | Fecha: Vie 20 Feb | 4pm - 6pm
#Nota: PC4 esta relacionada con PC3 — extiende la infraestructura previa



Contexto del Laboratorio

Objetivo : Ampliar la seguridad con monitoreo (Log Analytics),
cifrado de datos sensibles (Always Encrypted) y
almacenamiento de claves (Key Vault)

Parte 1 : WAF + Entra ID + App Services (mismo flujo PC3 con nuevos recursos)
Parte 2 : Log Analytics — monitoreo del WAF y autenticacion
Azure Storage — cuenta de almacenamiento
Key Vault — gestion de claves de cifrado
Azure SQL + Always Encrypted — cifrado de columnas DNI y Telefono

Tabla a proteger con cifrado:
Nombre — texto plano
Apellido — texto plano
DNI — CIFRADO (Always Encrypted)
Edad — texto plano
Direccion — texto plano
Telefono — CIFRADO (Always Encrypted)

Arquitectura — Flujo de Trabajo

```
Browser (Usuario MCaldas)
|
| HTTPS / TLS
v
Azure Application Gateway (WAF) — pc4-appgw IP: 52.228.126.148
|
| HTTPS + Log Analytics (pc4-waf-diagnostics)
v
```

```
Azure App Service — pc4-appservice-web
|
| OpenID Connect / OAuth 2.0
v
Microsoft Entra ID (Azure AD)

VM (admin-snet) --SSMS--> Azure SQL (pc4-sql-server) via Private Endpoint
Key Vault (pc4keyvault) -- Always Encrypted
```

PARTE 1 — ENTRA ID, APP SERVICES Y APPLICATION GATEWAY (WAF)

#El flujo es el mismo que PC3. Se usan nuevos nombres de recursos para PC4.

PASO 1 — Crear Usuario en Microsoft Entra ID

Azure Portal -> Microsoft Entra ID -> Usuarios -> + Nuevo usuario

usuario : MCaldas@emersonespinozasoutlook776.onmicrosoft.com
nombre : Manuel Caldas
contrasena : Uni1234%
nueva clave : Uni2026%

PASO 2 — Registrar Aplicacion en Entra ID + Client Secret

Microsoft Entra ID -> Registro de aplicaciones -> + New registration

Application : pc4-appservice

#Certificados y secretos -> + Nuevo secreto de cliente:
Value : Wq.8Q~NzRbs7SzTt6xw8U.kjIGjK6N1HRjupqbkS
ID Secreto : cd7ca861-7b7c-46ce-8f2a-1ef89c65a791

#ADVERTENCIA: guardar el value inmediatamente, no se puede recuperar.

PASO 3 — Crear Azure App Service

Azure Portal -> App Services -> + Crear

Nombre : pc4-appservice-web
Grupo : FIEE2026-1
Region : Canada Central
Plan : B1 (Basico)

PASO 4 — Configurar Autenticacion del App Service

App Service -> Autenticacion -> + Agregar proveedor -> Microsoft

Expiracion : 180 dias

#URL real del App Service (verificar tras crear):
pc4-appservice-web-dactgcg0dgdpa8c3.canadacentral-01.azurewebsites.net

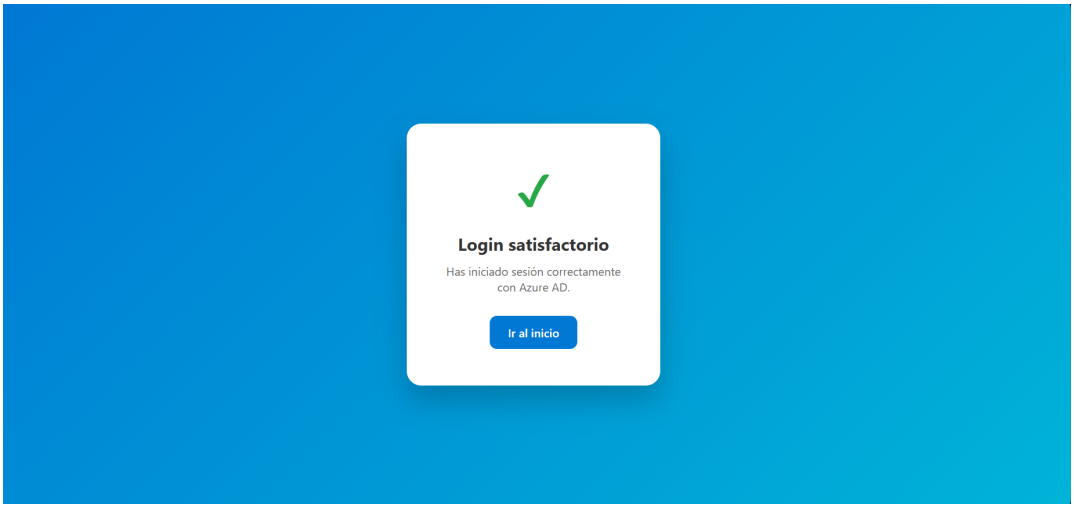
#Corregir URL de redireccion en Entra ID -> App -> Autenticacion:
https://pc4-appservice-web-dactgcg0dgdpa8c3.canadacentral-01.azurewebsites.net/.auth/login/aad/callback

PASO 5 — Insertar index.html en el App Service

App Services -> Herramientas de desarrollo -> Editor (Vista previa)
wwwroot/index.html

#Contenido del index.html (pagina de Login Satisfactorio):

```html
<!DOCTYPE html>
<html lang="es">
<head>
<meta charset="UTF-8">
<title>Login Satisfactorio</title>
<style>
body { height:100vh; background:linear-gradient(135deg,#0078D4,#00B4D8);
display:flex; justify-content:center; align-items:center; }
.card { background:white; padding:50px 40px; border-radius:20px;
box-shadow:0 20px 40px rgba(0,0,0,.2); text-align:center; }
.check { font-size:60px; color:#28a745; }
.btn { padding:12px 25px; background:#0078D4; color:white;
border-radius:10px; text-decoration:none; font-weight:600; }
</style>
</head>
<body>
<div class="card">
<div class="check">checkmark</div>
<h1>Login satisfactorio</h1>
<p>Has iniciado sesion correctamente con Azure AD.</p>
<a href="/" class="btn">Ir al inicio</a>
</div>
</body>
</html>
```



PASO 6 — Crear Application Gateway (WAF)

Azure Portal -> Crear recurso -> Application Gateway

Nombre : pc4-appgw
Grupo : FIEE2026-1
Region : Canada Central

VNet nueva : pc4-vnet
Subnet WAF : appgw-subnet — 10.0.1.0/24

Frontend : Publica — pc4-appgw-ip

Backend : pc4-backend -> App Service: pc4-appservice-web

Regla : pc4-rules | Prioridad: 100

```
Listener : pc4-listener
HTTP Settings : pc4-http-settings
IP Publica WAF: http://52.228.126.148
```

Seleccionar el nombre de host del destino de back-end

Reemplazar por un nombre de dominio específico

Nombre de host *

pc4-appservice-web-dactgcg0dgdpa8c3.canadacentral-01.azurewebsites.net

Sondeo personalizado

Usar sondeo personalizado

Sí No

Sondeo personalizado *

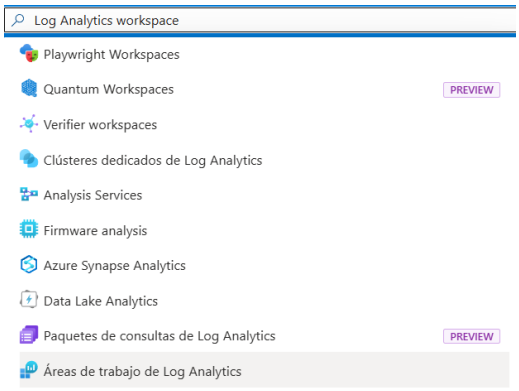
pc4-http-settings05722118-088d-45fc-ad63-5e3b176fa03e

PARTE 2 — LOG ANALYTICS — MONITOREO DEL WAF Y AUTENTICACION

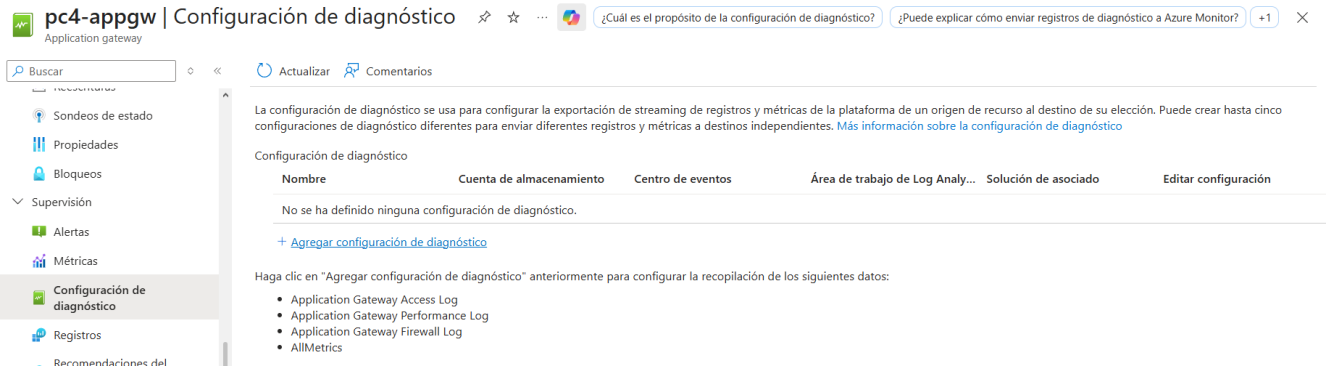
#Log Analytics permite ver requests al App Service, trafico del WAF
#y logs de autenticacion para medir que todo funcione correctamente.

PASO 7 — Vincular Log Analytics al Application Gateway

Azure Portal -> Areas de trabajo de Log Analytics

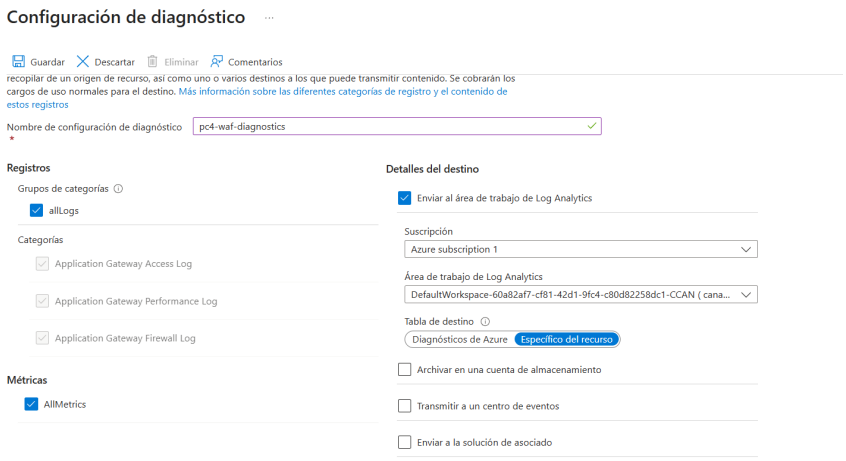


#Verificar que existe el workspace, luego vincular con el WAF creado:
Application Gateway -> Configuracion de diagnostico -> + Agregar



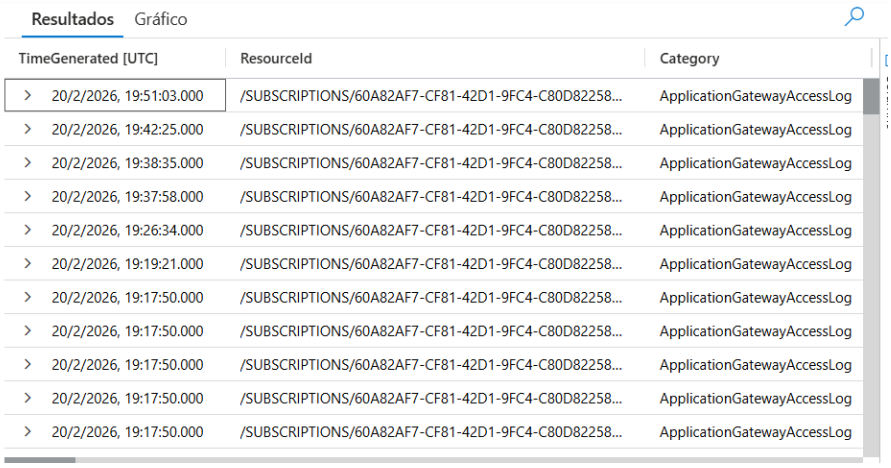
Nombre diagnostico : pc4-waf-diagnostics

#Marcar las categorias de log segun la imagen referencial:



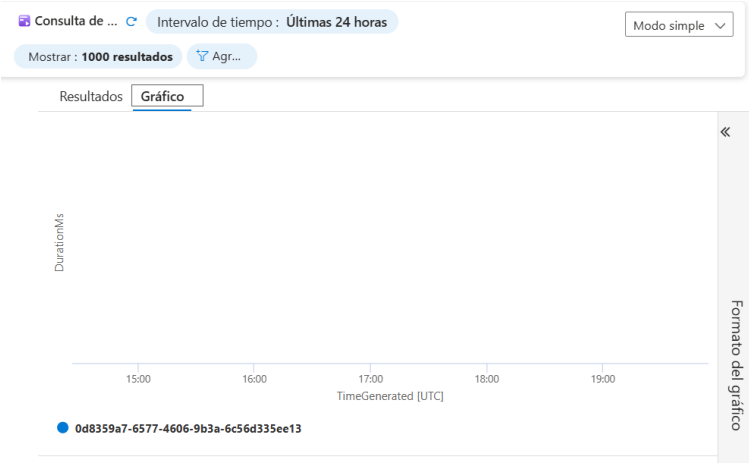
PASO 8 — Consultas en Log Analytics

```
#Ver todas las tablas disponibles:
search *
| summarize count() by $table

#Ver logs del firewall del Application Gateway:
AzureDiagnostics
| where Category == "ApplicationGatewayFirewallLog"
| sort by TimeGenerated desc
```



Resultados | Gráfico

TimeGenerated [UTC]	ResourceId	Category
20/2/2026, 19:51:03.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:42:25.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:38:35.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:37:58.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:26:34.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:19:21.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:17:50.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:17:50.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:17:50.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:17:50.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog
20/2/2026, 19:17:50.000	/SUBSCRIPTIONS/60A82AF7-CF81-42D1-9FC4-C80D82258...	ApplicationGatewayAccessLog



PASO 9 — Crear Cuenta de Azure Storage

#El Storage Account se usara como respaldo para Key Vault y cifrado

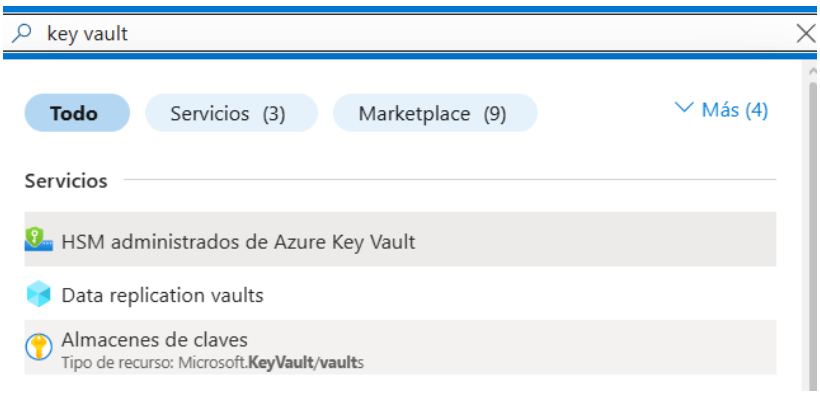
Azure Portal -> Storage accounts -> + Crear

Nombre : pc3storagewaf
Region : Canada Central
Redundancia : Local (LRS)

PASO 10 — Crear Azure Key Vault

#Key Vault almacena las claves maestras usadas por Always Encrypted

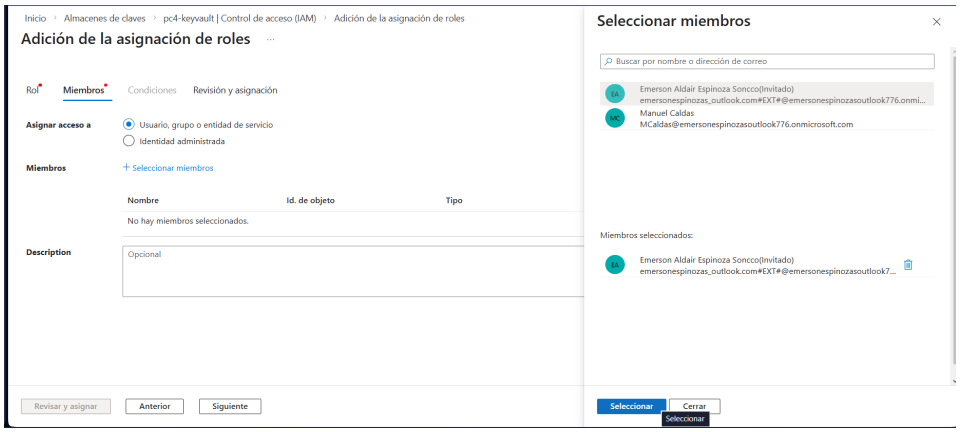
Azure Portal -> Almacenes de claves -> + Crear



Nombre : pc4keyvault
Region : Canada Central

Asignar Roles en Key Vault (IAM)

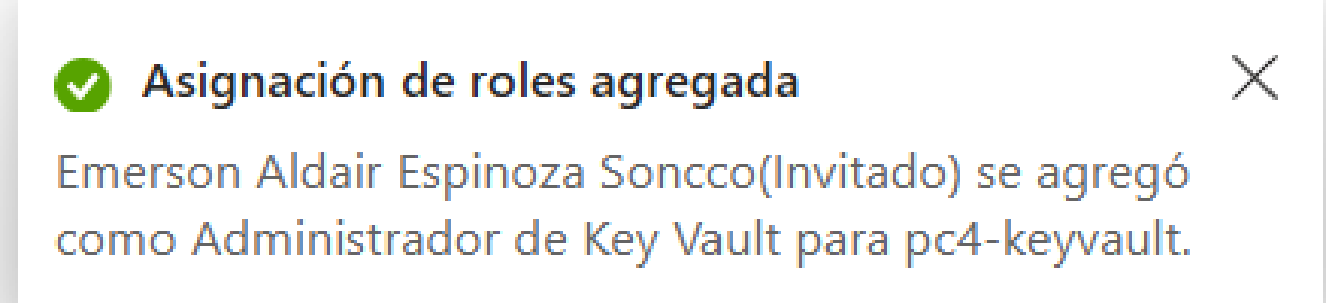
Key Vault -> Control de acceso (IAM) -> + Agregar -> Agregar asignacion de rol
Rol : Administrador de Key Vault



Administrador de Key Vault	Permite llevar a cabo todas las operaciones relativas a un plano de datos de un almacén de claves y a todos los objet...	BuiltInRole	Seguridad	Vista
Administrador de RBAC de Azure Kubernetes ...	Permite administrar todos los recursos de un clúster o espacio de nombres, excepto actualizar o eliminar espacios de...	BuiltInRole	Contenedores	Vista
Administrador de recursos de Azure Connect...	Puede leer, eliminar y volver a incorporar elementos de Azure Connected Machine, así como escribir en ellos.	BuiltInRole	Administración y gobe...	Vista
Administrador de seguridad	Rol de administrador de seguridad	BuiltInRole	Seguridad	Vista
Administrador de seguridad (heredado)	Este rol es heredado. En su lugar, use el Administrador de seguridad.	BuiltInRole	Seguridad	Vista

Revisar y asignar | Anterior | Siguiente | Comentarios

Asignaciones -> Agregar usuarios -> Revisar y asignar



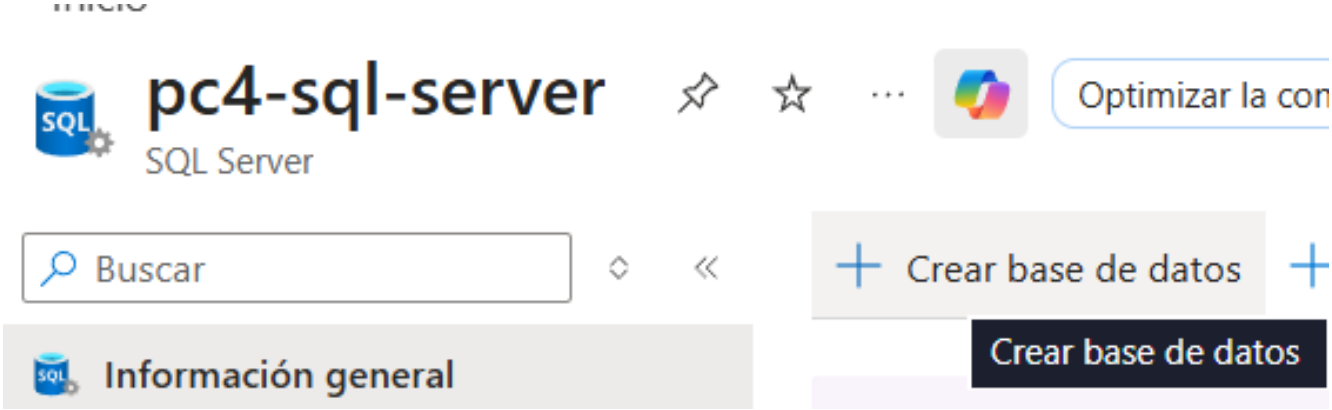
PASO 11 — Crear Azure SQL Server y Base de Datos

```
#Orden de creacion: SQL Server -> SQL Database -> Private Endpoint
```

11.1 — Crear SQL Logical Server

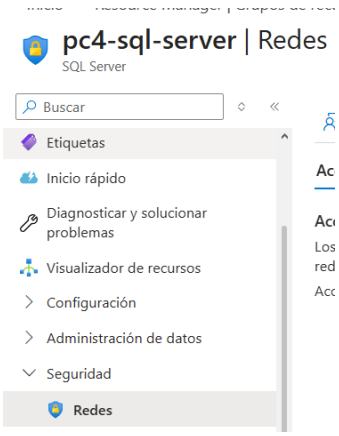
```
Azure Portal -> SQL Server (logical server) -> + Crear

Nombre : pc4-sql-server
Region : Canada Central
Autenticacion : SQL
User : sqladmin
Password : Adm!n1str@d0r#2026
```

11.2 — Crear SQL Database




```
Nombre : pc4-db-personal
Nivel : Basico
Almacenamiento: Local
```

PASO 12 — Configurar Private Endpoint para Azure SQL

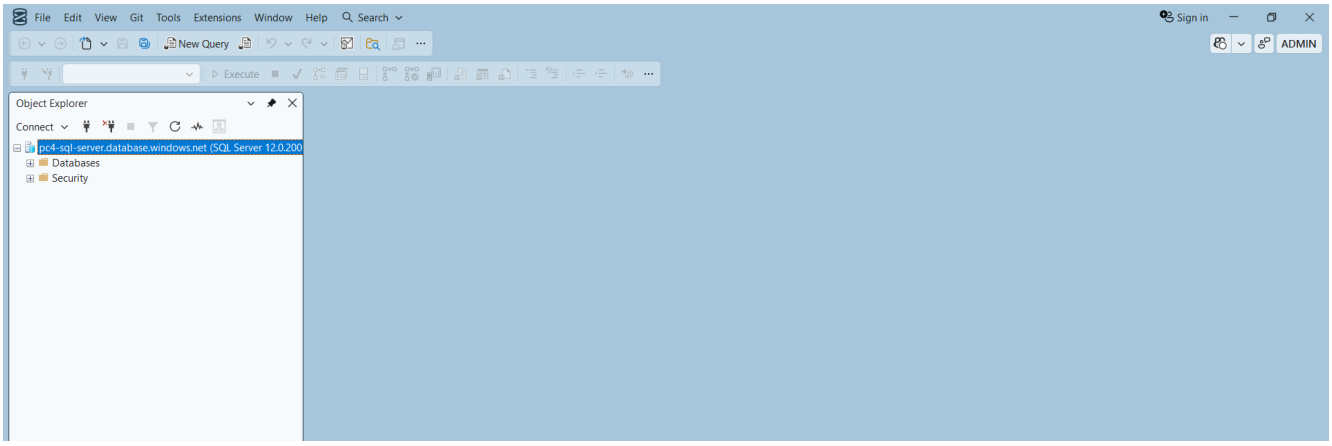



```
Azure SQL -> Redes -> Acceso Privado -> + Crear endpoint privado

Grupo : FIEE2026-1
Nombre : pc4-sql-pe
Region : Canada Central

#Estructura de subnets en pc4-vnet:
pc4-vnet
```

```
appgw-subnet <- WAF 10.0.1.0/24
pe-subnet <- Private Endpoint 10.0.2.0/24
admin-snet <- VM Administracion 10.0.3.0/24

Zona DNS : privatelink.database.windows.net
```

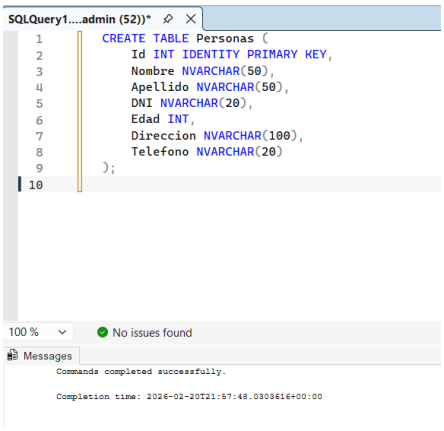
PASO 13 — Crear VM de Administracion y Conectar via SSMS

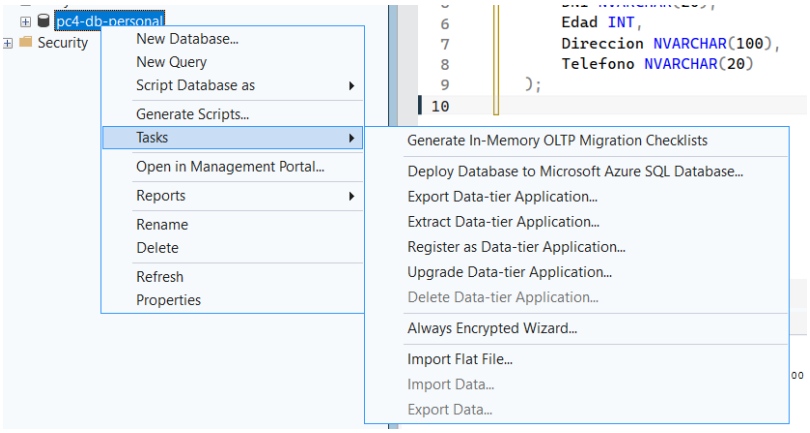
```
Crear VM en pc4-vnet -> admin-snet (10.0.3.0/24)
Region : Canada Central
User : Administrador
Password : Adm!n1str@d0r#2026

#Desde la VM, abrir SSMS y conectarse al SQL Server:
Server : pc4-sql-server.database.windows.net
User : sqladmin
Password : Adm!n1str@d0r#2026
```



PASO 14 — Crear Tabla Datos Personales en pc4-db-personal

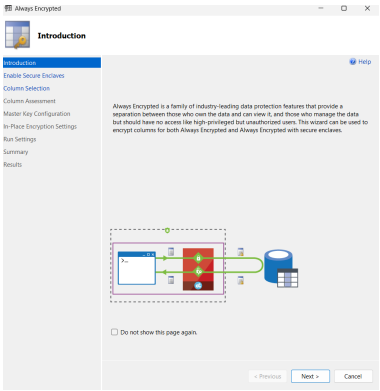
```
#Ejecutar desde SSMS conectado a pc4-sql-server
CREATE TABLE Personas (
Id INT IDENTITY PRIMARY KEY,
Nombre NVARCHAR(50),
Apellido NVARCHAR(50),
DNI NVARCHAR(20), -- sera CIFRADO
Edad INT,
Direccion NVARCHAR(100),
Telefono NVARCHAR(20) -- sera CIFRADO
);
```





PASO 15 — Cifrar Columnas con Always Encrypted Wizard

```
#En SSMS: clic derecho sobre la tabla Personas -> Always Encrypted Wizard
```



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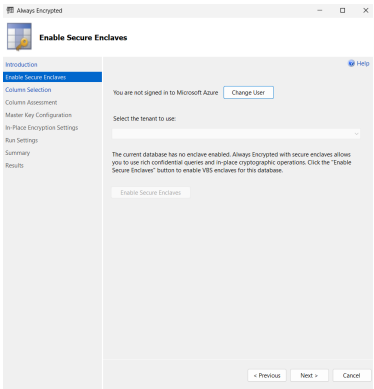

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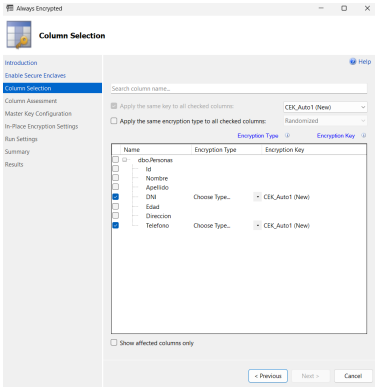
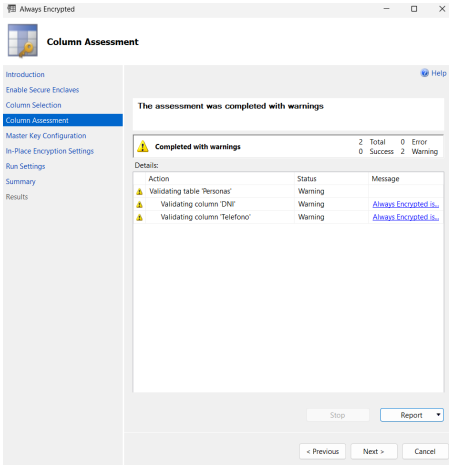


#Marcar las columnas DNI y Telefono para cifrar:



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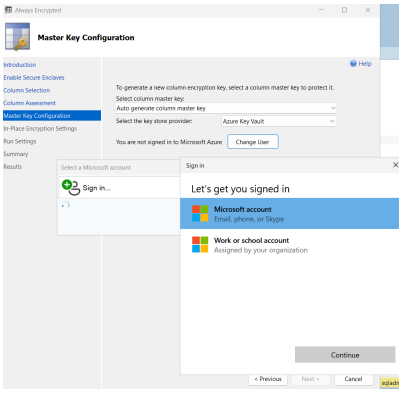
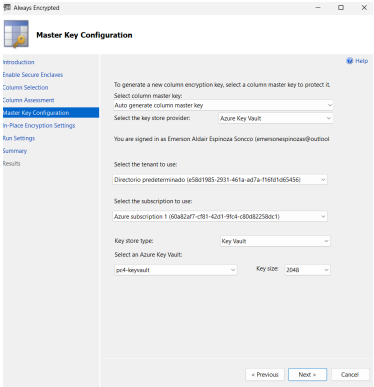


#Reiniciar la VM si hay problemas de conexion durante el wizard



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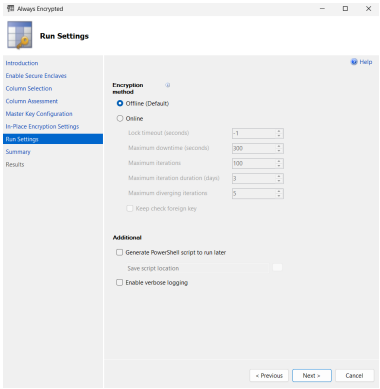


>Next -> Finish

Solucion de Error — 'We can't sign you in' (JavaScript bloqueado)

#Error: la VM de Windows bloquea JavaScript por IE Enhanced Security

1. Abrir Server Manager en la VM
2. Ve a Local Server
3. Buscar: IE Enhanced Security Configuration
4. Haz clic y configurar:

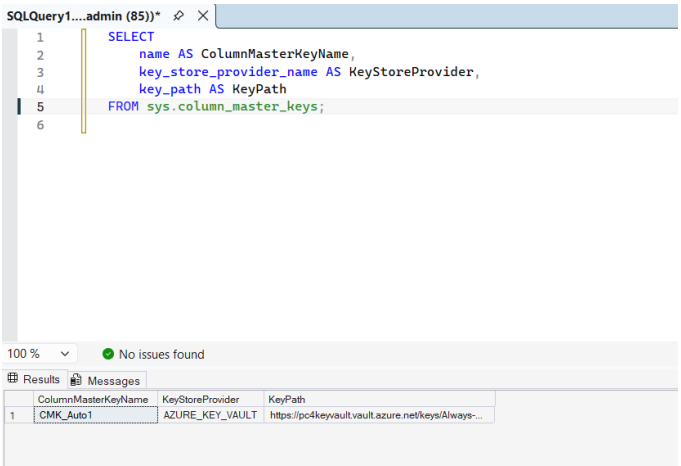
Administrators : OFF
Users : ON (puede quedar ON)

5. Cerrar SSMS
6. Volver a abrir SSMS

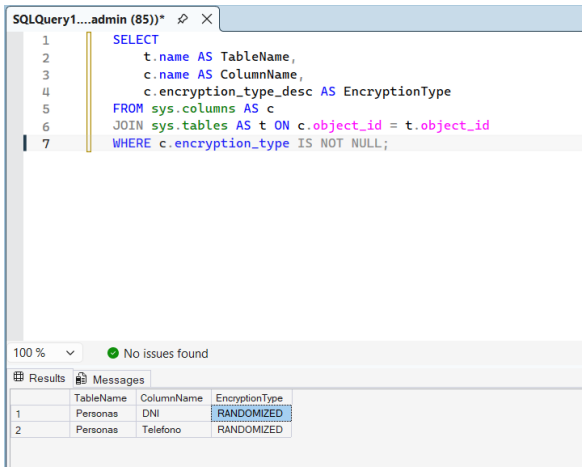
7. Ejecutar Always Encrypted Wizard nuevamente

PASO 16 — Verificar Datos Cifrados

```
#Ver las claves maestras de cifrado registradas:
SELECT
name AS ColumnMasterKeyName,
key_store_provider_name AS KeyStoreProvider,
key_path AS KeyPath
FROM sys.column_master_keys;
```




```
#Ver las columnas cifradas en la base de datos:
SELECT
t.name AS TableName,
c.name AS ColumnName,
c.encryption_type_desc AS EncryptionType
FROM sys.columns AS c
JOIN sys.tables AS t ON c.object_id = t.object_id
WHERE c.encryption_type IS NOT NULL;
```




```
#El resultado debe mostrar DNI y Telefono con EncryptionType = DETERMINISTIC
```